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Review

Beyond the public in controversies: A systematic review on social opposition and renewable energy actors

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ABSTRACT

Research attention for renewable energy controversies has been growing steadily. In studying such controversies, researchers have a tendency to focus on opponents, the public and local communities. There have been calls for some years now to broaden the scope of social science to include renewable energy technology (RET) actors such as project developers, governments or other actors involved in supporting or implementing renewable energy developments. Attention for RET actors is still limited and it is unclear where this area of research currently stands. Therefore, we present a systematic review of literature about RET actors in relation to controversies surrounding renewable energy technologies and infrastructure. The review of 89 publications shows two categories, consisting of eleven themes in total. The first category concerns RET actors' perceptions of the nature, impact and causes of public opposition, including specific groups of people and processes of engagement and decision-making. The second category describes different responses and strategies of RET actors to public opposition. Based on these findings, we present three observations about the current state of knowledge on RET actors and discuss implications for the broader field of renewable energy controversies. Future research should be directed at unravelling the diverse range of RET actors and their practices. This contributes to constructing a finer-grained understanding of the (development of) conflict dynamics between the actors engaged in controversies and action perspectives for the involved actors.

1. Introduction

Over the past decades, renewable energy controversies have received attention from scholars from multiple academic fields within social science [1], such as environmental psychology [2–4], sociology [5–8], ethics [9–12] and policy and planning [13–18]. In studying such controversies, researchers have a tendency to focus on opponents, the public and local communities [19,20]. This is especially the case for the topic of social acceptance of renewable energy (See [21–27] for reviews). This tendency borders "obsession" [28] and has resulted in the neglect of other key actors in the energy domain relevant for understanding controversies [3,19].

This predominant focus on specific actors and neglect of others is problematic due to the complex nature of controversies. In this type of conflict, a multitude of different public and private actors with various roles, representing different, potentially conflicting interests interact [17,18,29–31]. Aside from the general public, local residents or communities opposing a project, there are other key actors involved in controversies which are referred to as 'renewable energy technology' (RET) actors [17]. Walker et al. describe RET actors as "a broad category of people in organizations with roles in supporting or implementing [renewable energy technology] developments – including developers, consultants, PR and marketing companies, trade associations, financiers and technology manufacturers" [[17], pp. 4–5].

Interactions between RET actors and opponents are "potentially significant in shaping how the dynamics of local responses evolve and how different actors learn, react and strategically behave in relation to each other" [[17], p. 5]. In these interactions, it is not just the choice of a particular renewable energy technology that can trigger or influence a controversy: processes of public participation and formal decision-making on renewable energy can also incite public opposition [17,32,33]. For example, processes based on Not In My BackYard (NIMBY) assumptions often leave local stakeholders feeling powerless and disillusioned by formal processes of engagement [34]. Another

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example is the "decide-announce-defend" (DAD) model: "when results developed by 'educated experts', project developers or government are simply communicated to the public" [[35], p. 252]. Both NIMBY assumptions and DAD approaches can result in (increased) public opposition [33,36,37]. This means that decisions and choices, and underlying assumptions, made by RET actors on how to engage with public opposition can teach us valuable insights about (the development of) controversies and related dynamics of interaction [17].

Better understanding of controversies is crucial as many have led to delays or cancellations of renewable energy projects [33,38–40]. However, such an understanding cannot be based on research into only the public, local residents or those opposing renewable energy projects. The complex nature of controversies also requires in-depth investigation into and understanding of RET actors, of the roles they play and their assumptions and expectations underlying decisions and actions [17,41–43]. Such insights create opportunities for researchers to provide diverse and concrete recommendations for inclusive decision-making in energy controversies, such as recommendations that are more tailored to the challenges RET actors face in this area.

There have been calls for some years now to broaden the scope of social science research to include RET actors [19,42,44,45]. Nevertheless, attention for RET actors still seems to be limited and it is unclear where this area of research currently stands. Such insight is necessary for developing a substantiated research agenda. Therefore, we present a systematic literature review and ask the question: What do we know of RET actors when it comes to controversies surrounding renewable energy technologies and infrastructure? Our review is restricted to studies that cover energy technologies for production and transport of renewable energy. This includes technologies like wind, solar and hydropower, as well as the (re)development of infrastructure, such as transmission lines that are necessary for energy system transformation.

Below, we will first discuss the method we used for this literature review. Then, in Section 3, we present our findings. We identified two categories in existing social science research: (1) RET actors' perceptions of public opposition and (2) RET actors' responses to public opposition. Next, in Section 4, we discuss three observations related to the nature of the generated insights and characteristics of research approaches of the reviewed publications. In our conclusion (Section 5), we make a case for future social science research to be more explanatory and reflective of the diversity of RET actors.

2. Method

2.1. Data collection

We conducted a systematic literature review which uses an "explicit, and reproducible method for identifying, evaluating, and synthesizing" a body of academic literature [[46], p. 3]. An iteratively developed list of key words was used to identify relevant records in academic databases Scopus and Web of Science.[1] The search string consisted of four word groups with variations of key words that together formed the building blocks for the scope of our review. We focused on the *development* (category 1) of *renewable energy technology and infrastructure* (category 2) by (*groups of*) *actors* (category 3), with a specific focus on projects that are characterized *by public opposition* (category 4, also see Table 1). For categories 1 and 2, the selection of key words was a straightforward process:

- Category 1 included key words referring to the development and implementation of specific projects regarding renewable energy technology or infrastructure. This meant that literature that concerned the operation or dismantling of energy technology was excluded from the search.
- Category 2, 'Renewable energy technology and infrastructure', included key words reflecting types of renewable energy technology and infrastructure.
- For the third category, we used the description of RET actors from Walker et al. [17] as a starting point for formulating the key words (see Section 1). The terms 'technology promoter', 'PR company' and 'marketing company' did not result in any hits with test rounds in Scopus so these were omitted as key words. 'Trade association' also resulted in zero hits so we changed this to 'association'. 'Project manager', 'initiator' and 'cooperative' were added based on publications that were deemed relevant for this review (including [43,47–50]). Also key words referring to governmental actors were added to the list, as these actors are often involved with supporting or implementing renewable energy developments [17]. Test searches on Scopus with the variations 'parish', 'civil servant' and 'housing corporation' led to zero results so these were not included.
- 'Public opposition' refers to societal, publicly salient conflict, opposition or controversy [51] in the context of renewable energy projects and infrastructure. The key words in the category 'public opposition' were variations on key terms that are commonly used in academic literature when discussing public opposition to renewable energy technology implementation. A test run revealed that more generic conflict-related words, such as 'contest', 'appeal', 'refuse', and 'reject', resulted in too many irrelevant results, so these were left out.

We performed our search in the online databases of Scopus and Web

Table 1
Overview of key words used in search string. The * was used to include variations of key words.

Category 1: project development	Category 2: renewable energy technology and infrastructure	Category 3: RET actor		Category 4: public opposition
Siting	Renewable energy	Developer	Policy maker	Controversy
Planning	Energy project	Initiator	Politician	Conflict
Develop*	Energy technology	Industry	Government	Opposition
Implement*	Low*carbon energy	Entrepreneur	Municipality	Hindrance
Deploy*	Wind	Consultant	Province	Contestation
	Solar	Corporation	Canton	Contention
	Photovoltaic	Association	County	Protest
	Biomass	Manufacturer	State	Resistance
	Bioenergy	Cooperative	Policy actor	
	Tidal	Compan*	Decision-maker	
	(Ocean) Wave energy	Energy actor	Protagonist	
	Hydro*power	Key actor	Supporter	
	Geothermal	Stakeholder	Business	
	Nuclear energy	Electric utility	Grid operator	
	Hydrogen			
	Energy infrastructure			
	Transmission line			
	Power line			
	High power voltage			
	Pipeline (for hydrogen or biogas)			

[1] In this process, we used key words from several articles on RET actors, including [19,43,44,95], to further develop the list of key words.

of Science (January 27th 2021). We limited our review to records in English, which is a common practice for this type of literature review due to replicability and translation-related practicalities [52].[2] To be sure of scientific rigor, we only included records from international, peer-reviewed academic journals and established publishers [53]. The combined searches resulted in 3370 records, which were stored in an Excel database.[3]

In the following step, records were screened using their title and abstract. The eligibility of a record was determined based on the following criteria (reflecting the four building blocks of the scope for this reviews): it contains empirical work on (1) actors in the process of (2) implementing (3) renewable energy projects or infrastructure which are characterized by (4) public opposition.

Results that did not address all four building blocks were excluded from the review, such as those publications that:

- Do not concern public opposition, but address conflict between governmental actors or between private actors;
- Strictly focus on other perspectives than those of RET actors (communities, opponents, etc.) related to the planning and development of a certain renewable energy technology or project;
- Do not concern the development or implementation of a specific project; or
- Do not specifically report original empirical research.

For several articles, the abstracts met all the requirements, but screening of the full text showed otherwise. This was often the case for results focusing on public or community acceptance and the effects of participation. These results were only included if the processes did take place in the context of public opposition and included for example perspectives, rationales or discourses of RET actors. If this was not the case, those results were excluded. Reviews of previous empirical work were also excluded. Work on energy policy evaluation was not included if it did not concern the implementation of energy projects or infrastructure. Also excluded were records strictly focusing on technical or financial aspects (see Appendix A for a flow chart of this process). This resulted in a final set of 89 records being included in the review (see Table 2 for an overview per theme and Appendix B for a full overview of the included publications).[4]

2.2. Analysis of publications included in review

Next, the full texts of all publications included in the review were coded using ATLAS.ti (version 9.1.4.0). We used an inductive coding approach to allow bottom-up generation of codes [54], as the goal was to create an overview of social science literature on RET actors, without limitations from theoretical frameworks. Although our coding approach was inductive, it was sensitized by our main research question. As such, our coding was guided by the following questions: how do RET actors think about public opposition? How do they engage with or respond to public opposition? Why do they engage with public opposition in a particular way? In addition, we allowed other topics to emerge as part of the bottom-up approach and assigned codes concerning investigated actors, renewable energy technology or infrastructure and geography. Once all publications had been coded by the first author, the inductively generated codes were clustered into themes. The codes and themes were subsequently validated in several discussion rounds between the two authors. Two main categories emerged after this step:

- Perceptions of public opposition among RET actors
- Responses of RET actors to public opposition

Most records were assigned to one of the eleven themes, some (N = 25) were assigned to multiple themes. The results are presented in Section 3 (See Appendix C for an overview of the codes per theme and category).

Table 2
Overview of the number of publications per theme.

	Theme	Publications
Category 1: RET actors' perceptions of public opposition	The impact of public opposition according to RET actors (Section 3.1.1)	5
	Causes of public opposition according to RET actors: processes of participation and decision-making (Section 3.1.2)	11
	Causes of public opposition according to RET actors: specific groups of people (Section 3.1.3)	19
Category 2: RET actors' responses to public opposition	Using public engagement instrumentally to prevent and reduce opposition (Section 3.2.1)	17
	Taking a reactive approach to public opposition and focus on project development (Section 3.2.2)	6
	Making strategic choices in public engagement to prevent public opposition (Section 3.2.3)	6
	Educating the public to reduce opposition (Section 3.2.4)	11
	Using community benefits to reduce public opposition (Section 3.2.5)	12
	Contrasting claims to delegitimize public opposition (Section 3.2.6)	14
	Using regulatory and power structures to restrict public opposition (Section 3.2.7)	14
	Accommodating public concerns to reduce opposition (Section 3.2.8)	13

3. Results

Figs. 1–4 provide overviews of the focus of the publications included in this review in terms of types of actors, renewable energy technologies & infrastructure, geography and sources of publications. Fig. 1 shows the actors studied in the publications. The category 'Multiple actors, including RET actors' concerns publications that study the interplay between a multitude of actors involved in a renewable energy controversy including RET actors. Many of the publications in this category only briefly or superficially discuss RET actors as part of a wider array of actors. In several publications there are no more than a couple of sentences specifically on RET actors, for example [48,55,56]. The high number for this category forms a striking contrast with the number of publications that focus on sub-categories of RET actors. Fig. 2 shows the renewable energy technologies and infrastructure that publications focus on. More than 50 % of the included publications concern onshore wind (N = 48), with in total 60 publications for wind. The U.K., Mexico and Canada are the geographical areas that are most often investigated (Fig. 3). For Canada, 6 out of 8 articles have a single focus on Ontario. Fourteen countries were investigated once. Fig. 4 shows the sources of the publications: fifty different journal titles, ten of which published more than one article. Most articles are published in journals with a focus on policy and planning, where Energy Policy has the highest number of articles. All publications (with the exception of one) were published between 2001 and 2021.

We identified two general categories in this literature review, each

[2] This is a pragmatic choice, as both authors are proficient only in English and Dutch. Though pragmatic, this methodological choice impacts the publications that are taken into account for this literature review. We are aware that we are potentially missing relevant publications in languages other than English.

[3] Not related subject areas, such as Physics, Material Sciences, Agricultural and Biological Sciences, were excluded in both databases.

[4] Two publications [149,150] were excluded from the review due to issues with institutional access.

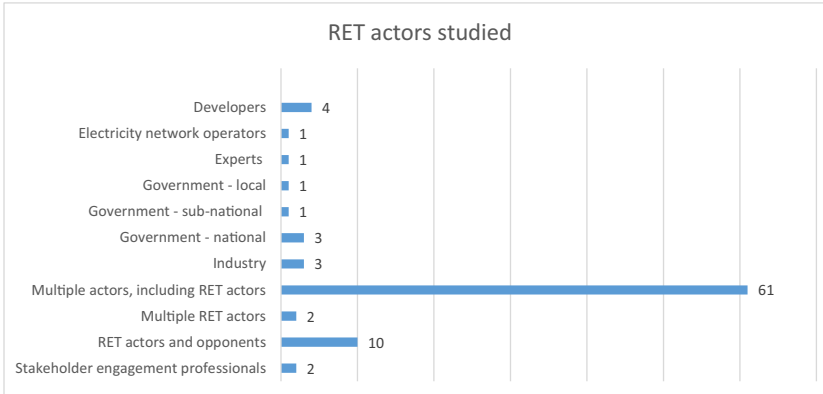

Fig. 1. RET actors studied in empirical social science research (in alphabetical order). 'RET actors and opponents' concerns publications that analyze conflicts from two perspectives: RET actors and opponents. 'Multiple actors, including RET actors' concerns publications that study a multitude of actors involved in a renewable energy controversy, including RET actors.

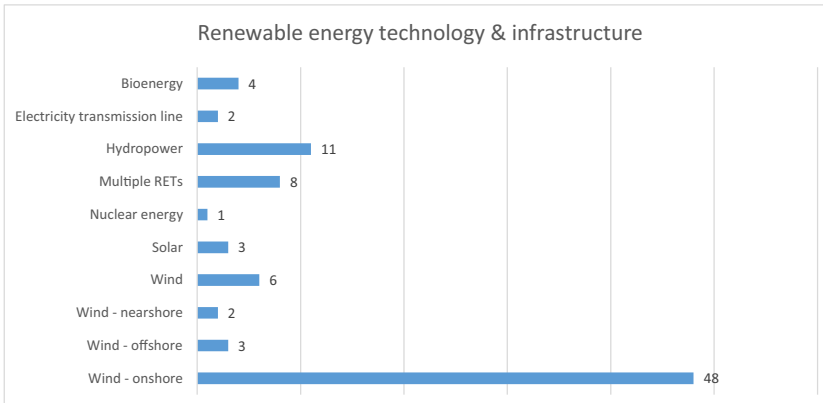

Fig. 2. Renewable energy technologies and infrastructure studied in empirical social science research on RET actors (in alphabetical order). The general code 'wind' was used for results that did not clearly specify nearshore, offshore or onshore wind.

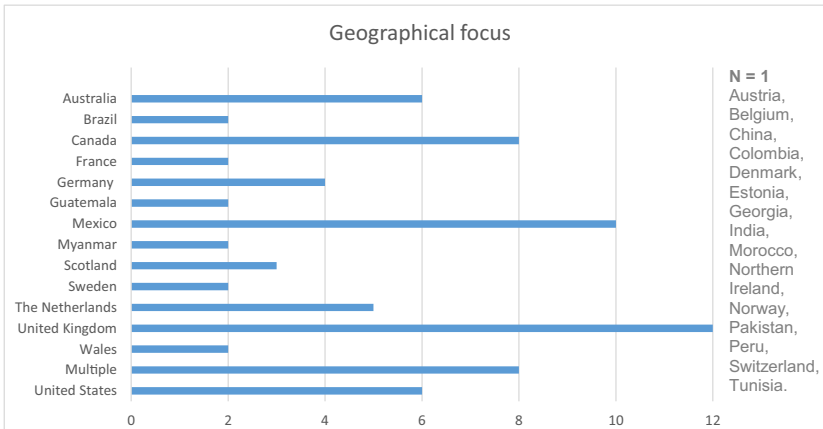

Fig. 3. Geographical focus of empirical social science research on RET actors included in review (in alphabetical order).

consisting of several themes (eleven in total, see Table 2). The first category discusses publications that report on RET actors and their perceptions of public opposition, including ideas about the nature, causes and impact of public opposition (Section 3.1). The second category concerns publications that address how RET actors respond to public opposition (Section 3.2). Below, each theme is described and illustrated by several examples (see Appendix A for a full overview of publications per theme).

3.1. RET actors' perceptions of public opposition

Below we discuss the first category of this literature review. This category reports on three themes distilled from the included publications on RET actors and their perceptions of public opposition, including

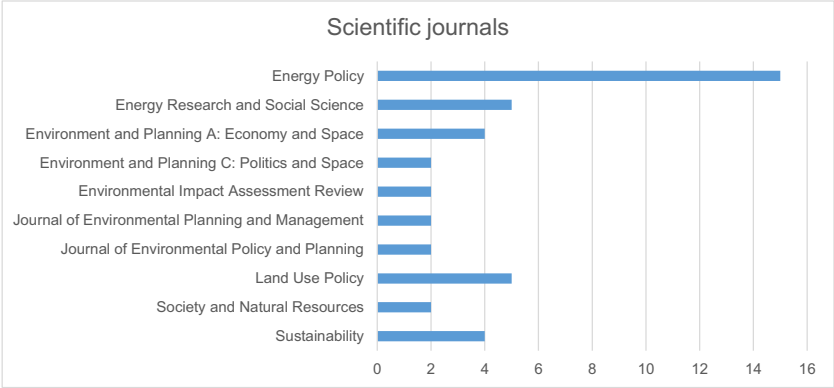

Fig. 4. Titles of journals containing publications included in this literature review (in alphabetical order). Journal titles with $N = 1$ and book chapters ($N = 7$) are not included in figure.

ideas about the nature, impact and causes of public opposition.

3.1.1. *The impact of public opposition according to RET actors*

The literature reports on different assumptions and discourses of RET actors with regards to public opposition. Most RET actors have negative assumptions: public opposition is perceived as a barrier to the development of renewable energy projects [43,48,57,58]. A small number of publications report on RET actors who perceive opposition as neutral or positive. For instance, some developers find wind energy resistance movements to have a positive impact on a development, as they increase developers' accountability and "in some cases [secure] other benefits for the community that would not have been there otherwise" [[43], p. 7]. Walker et al. [58] show how industry and policy actors construct imaginaries of the public from first-hand and mediated experience and knowledge; they describe the impact of these imaginaries as a "shared expectation of an ever present latent but conditional public hostility to renewable energy project development [that] is seen as shaping the material forms of the technologies, their evolving spatiality, and practices of public engagement involved in obtaining project consent" [p. 931]. Cass and Walker [59] find both neutral and negative perspectives on opponents of renewable energy developments among developers and other industry actors: the negative perspective views opponents as highly emotional, unreasonable and illegitimate, while the neutral perspective views such responses "as understandable and an inherent part of the politics around proposals for developments such as wind farms" [p. 68]. The authors find that, despite the different perspectives, developers perceive emotions of opponents as needing to be managed to keep the decision-making process "unemotional, objective and rational" [p. 68].

3.1.2. *Causes of public opposition according to RET actors: processes of participation and decision-making*

RET actors see different causes for public opposition, several of which are related to processes of public participation, spatial proximity to the project, governmental authority in decision-making and type of technology [36,59–65]. RET actors feel that causes related to community engagement and public participation play a particularly important part in (triggering) public opposition. According to a survey among representatives of national wind energy associations and experts in Europe, inadequate public participation is a reason for opposition to wind energy projects. At the local level, for example, "the less influence actors have the more likely it is that problems with social justice, participation and conflictive situations can arise" [[66], p. 321]. Juerges and Newig [67] show that German wind energy developers think less opposition will arise if decisions are made at the local governmental level, as opposed to top-down planning. Representatives of a Brazilian regulatory agency and electricity company state that not organizing community engagement and disseminating information on possible benefits can trigger local opposition when developing large-scale solar projects [68].

3.1.3. *Causes of public opposition according to RET actors: specific groups of people*

RET actors, specifically wind energy developers, industry representatives (including a cooperative) and local government officials, think in many cases that public opposition originates only from a small group of people [19,43,48,58,69–74]. Both local governments and developers see small groups of opponents as a particular challenge when balancing different local interests [72]. These small groups of vocal opponents are often perceived as coming from outside of host communities and stirring up local opposition [43,67]. Multiple articles elaborate on this [19,43,58,69,70,72] and show that RET actors refer to the majority of residents as supportive, to explicitly contrast that to the image of small vocal groups of opponents. RET actors perceive this majority as silent or apathetic, not actively expressing any type of opinion: "Passive local people are often described by authorities and developers as lacking interest in the process, or as a majority quietly in support of the project - acceptance is often measured in terms of activity" [[70], p. 700]. Thus, public silence is seen as equal to acceptance [70].

NIMBYism and knowledge deficits are also common causes of public opposition according to RET actors [75]. Although few UK developers explicitly label opponents as NIMBYs, "self-interest was often considered to be at the root of opposition" [[19], p. 250]. Scherhaufer et al. [76] also report on public opposition explained as driven by self-interest, specifically in the case of environmental impact assessment (EIA) procedures. In addition to the NIMBY label UK developers also ascribed knowledge deficits to opponents: "[they] were not so much characterized as having *insufficient* knowledge as having *incorrect* knowledge. Developers tended to characterize opponents as having faulty knowledge, rather than simply lacking information" [[19], p. 251]. Though there is antipathy toward the use of the NIMBY model among National Grid representatives in the U.K. [65], most publications concern examples of discourses revolving around the attribution of the NIMBY label and knowledge deficits to those opposing developments. These publications report on wind energy developers and industry actors, including consultants, marketing and public relations companies [19,64,73,77–79]. Related discourses are also found in the institutional context, in policies regarding wind power implementation and community engagement [37,38].

3.2. *RET actors' responses to public opposition*

The second category concerns publications that address how RET actors respond to or interact with public opposition.

3.2.1. Using public engagement instrumentally to prevent and reduce opposition

Several articles describe how RET actors deploy participation instrumentally to prevent public opposition and gain acceptance [36,58,64,74,80–88]. For example, Walker et al. [58] identify highly instrumental motivations among RET actors to organize public engagement, with most focusing on "obtaining planning permission, speeding up decision processes, and minimizing complications because of an anticipation that these could arise from antagonistic public responses" [p. 941]. Anderson [89] reports on similar motivations in a public engagement process for an Australian wind farm which was designed to "reduce conflict and achieve acceptance" [p. 106]. Although publications that report on instrumental motivations are in the majority, there are a few that go beyond a mere instrumental point of view on participation [44,49,50]. These publications report on RET actors who perceive early and active engagement as good practice and opposition as a potential source of information. The investigated RET actors think that participation increases the chances of successful implementation and helps with reducing opposition.

3.2.2. Taking a reactive approach to public opposition and focus on project development

While the previous theme concerned an instrumental and preventive response to public opposition, this theme concerns RET actors who have a reactive approach to public opposition. Both Pasqualetti and Schwartz [90] and Pepermans and Loots [6] report on DAD-like approaches in solar and wind siting respectively, where opposition was responded to by defending announced decisions. In these cases, informing and consulting local stakeholders took place once project plans were at an advanced stage of formal decision making, leaving little room for addressing local concerns. Jami and Walsh [91] report on similar findings in five wind projects, characterizing these processes as "quite a reactive and tardy public consultation performance" [p. 15]. Van de Grift et al. [44] identified wind developers who had a similar approach. These developers strictly focus on technical project development and, as such, they feel they are not responsible for dealing with public opposition; they view opponents as outside of their sphere of influence and choose to not actively engage with them, but rather strictly adhere to legal requirements for public engagement. Another explanation for a reactive approach to public engagement and strictly adhering to regulations by RET actors are vague and unclear regulations regarding public participation and specifically the Social Impact Assessment (SIA) [92,93].

3.2.3. Making strategic choices in public engagement to prevent public opposition

In order to prevent public opposition, RET actors make strategic choices regarding the project, type of technology and planning policy. For example, they apply specific locational strategies, such as "the reuse of existing or historic locations of [renewable energy technologies] for the installation of newer technology", as less disruption and opposition is anticipated compared to a new location [[58], p. 940]. In anticipation of public opposition related to wind energy, a Dutch energy cooperative made the decision to focus on solar projects instead, assuming general public support for this technology [71]. This strategic choice of technology served to build trust and support for the cooperative's cause and avoid public opposition.

Two authors report on governmental actors making changes to policy for planning of renewable energy in anticipation of public opposition. Bues [95] provides an example of spatial-discursive strategies from sub-national governments: "In anticipation of conflict, Ontario [Canada] has chosen to raise planning responsibility for renewable energy facilities to the provincial level. In contrast, Brandenburg [Germany] has chosen to endorse its regional planning approach" [p. 34]. Martinez and Komendantova [92] describe how the Mexican federal government introduced social impact assessments (SIA) in laws and regulations in response to ongoing public opposition, hoping this would provide 'an institutional mechanism' to deal with controversies on social impacts [p. 3].

While RET actors were aware of the limited public participation in the projects investigated by Pepermans & Loots [6], "they feared that more participation would only lead to more disappointment and increasing numbers of protests" [p. 324]. Although they knew about the potential positive effects of active public engagement, they "did not see many options on how to further engage communities. They blamed the scarcity of possible locations, the competitive situation and the strict spatial planning guidelines" [p. 324]. A similar sentiment is found among Austrian developers of wind energy, for whom the use of public opinion polls is controversial: although they find this tool can increase the accountability and legitimacy of decision-making, developers also perceive it as creating potential for polarization among local communities [76].

3.2.4. Educating the public to reduce opposition

This RET actor response is about educating the public to reduce opposition. RET actors believe that educating the public on technical aspects related to risks and safety is an appropriate response to public opposition: by providing what they believe to be objective information, they attempt to debunk the "faulty" assumptions of opponents on wind power. This way they hope to convince opponents of the benefits of wind energy, and overcome NIMBY sentiments and knowledge deficits [6,59,70]. Burningham et al. elaborate on this:

> Developers speak of their hope that the deficits of understanding, information, and experience they identify might be filled by education or the provision of appropriate "experience", thus potentially transforming opponents into supporters. Here a clear public deficit model of understanding is evident, with the assumption being that given "facts" or shown "the reality", members of the public will think more like the experts. (…) [[19], p. 257–258].

Worldwide, wind energy developers' legitimize their projects by referring to scientific consensus on climate change and surveys that highlight general public support for wind energy, a response that reveals NIMBY assumptions [6,19,59,73,96]. In the case of the Australian government, Hindmarsh [37] found NIMBY assumptions leading to "a regressive approach to community engagement (…) to facilitate wind farm development informed by an 'education' programme that dismally reflects the widely criticized information deficit approach (…)" [p. 559].

The belief that educating the public on the benefits and safety of energy technologies will reduce public opposition is also found among developers of solar, nuclear and biomass projects [57,93,97]. Developers with what is described as a TINA (There Is No Alternative) attitude to a proposed project are skeptical of social and environmental public concerns and expect techno-centric arguments will convince opponents [[97], p. 793].

3.2.5. Using community benefits to reduce public opposition

RET actors often use community benefits as a way to reduce public opposition [64,98–103]. Developers explain their increased attention for community benefits as fueled by the challenge of obtaining community consent [[104], p. 548]. For instance, according to Canadian wind energy developers, "financial compensation is needed in order to address the situation where landowners holding turbine leases tend to be the only beneficiaries (…)" [[105], p. 760]. Walker et al. [58] identify a general utilitarian motivation for introducing community benefits in renewable energy projects among U.K. developers, who perceive public opposition as being driven by a balance between costs and benefits: "If this balance could be redressed through a monetary transaction, then anticipated opposition could be counteracted and potentially overcome" [p. 942]. Cowell et al. [104] find instrumental motivations for the use of community benefits in wind energy developments in Wales.

For some policymakers, community benefits are part of conflict

resolution: they perceive "'improving' the provision of community benefits from wind farm developers as one way of resolving social conflicts around the siting of facilities, and thereby expediting expansion" [[104], p. 544–545]. When developers use community benefits to address anti-wind sentiments, they feel that such benefits in the form of financial compensation can trigger jealousy [105]. Financial compensation being perceived as bribery is another fear expressed by Canadian wind energy developers: "most [developers] struggled with the idea of making direct payments to host communities for two main reasons: (1) perceptions of bribery and (2) defining the spatial scope within which to distribute payments" [[43], p. 9]. For the Australian wind industry community benefits are a way to "secure community support and reduce (the impacts from) community opposition" [[80], p. 125]. Financial compensation is often an explicit part of processes aimed at gaining community consent or a social licence to operate (SLO) [74]. So called 'Community Engagement Plans' are an example, which "encapsulate the intentions of developers for community engagement and benefit-sharing in specific wind farm developments" [[80], p.125]. In SLO processes, negotiation on financial compensation on community funds is common. SLO processes "are set up beyond compliance by most wind farm developers. One [developer] described the motivation for this as making a positive contribution to the region beyond lease payments to several landowners" [[74], p. 455].

3.2.6. Contrasting claims to delegitimize public opposition

This response concerns framing practices of RET actors in the face of public opposition with the purpose of delegitimization. Some RET actors have framed landscape in a neutral manner, which denies landscape constructs by residents [106]. In the case of resistance to a Dutch wind farm, the national government simply denied allegations of participatory tokenism and did not engage in further discussion, aside from a brief statement which declared that "the whole process had been open and participatory" [[13], p.16]. In addition to this, two specific ways of RET actors contrasting claims in an attempt to delegitimize public opposition can be observed in the literature: (1) contrasting local or personal interests with national/global interests, using sustainability and the NIMBY label, and (2) contrasting experts with laymen.

With respect to the first, multiple authors have observed RET actors using sustainability as a contrasting frame in their communication to delegitimize opposition [73,76,107–109]. Sustainability was one of several frames used by the Brazilian national government in a controversial hydropower project "to deflect opposition criticism and widen the scheme's perceived beneficiaries" [[108], p. 276]. Wind developers in Mexico used similar contrasting framing: "balancing local concerns of damage against the global gains of climatological cleansing has been a critical element in positioning the wind park and the company itself as environmental friendly" [[107], p. 384]. For these developers, presenting their practice as 'environmentally upstanding' [p. 396] was a way to delegitimize opposition. In one case of opposition to solar developments, the Moroccan government took on a 'technomanagerial eco-consensus' frame to contrast and overrule the complaints of residents, stating that "the benefits of solar power outweighed any countervailing concerns: popular demands represented a barrier to clean energy to be overcome by transforming political claims into a technocratic problem to be addressed through development interventions" [[110], p. 542].

Several researchers identified RET actors using the NIMBY label to delegitimize public opposition [19,37]. According to Walker et al. [58], developers framed local concerns regarding the impact of proposed wind farms as disingenuous, actively negating opponents' concerns. Tafon [111] reports on an offshore wind developer painting a picture of "recalcitrant NIMBYs, who would do everything in their power to block development, principally on the grounds of their limited self-interests, which would be detrimental to the interests of the 'broader' community" [p. 170]. Some RET actors highlight self-interest by vilifying residents opposing wind farms: "politicians and entrepreneurs [refer to opponents] as a 'minority' of 'lazy,' 'drunk,' and 'violent' 'bandits' that are afraid of change and are just looking to get a better contract with companies" [[112], p. 943].

The second manner in which RET actors delegitimize opposition is by contrasting experts (or expert knowledge) with laymen (or lay knowledge). In the case of a hydropower project in Pakistan, Niazi [113] reports on the use of the knowledge deficit label: "the state and its administrative agencies that support dam building rest their case on 'expert' knowledge, while dismissing the counter knowledge produced by the anti-dam coalition of environmentalists, nationalists, and spiritualists as 'inexpert'." [p. 441] By presenting their own arguments as objective and true, developers contrast opponents as unknowledgeable and thus attempt to delegitimize public opposition. Supporters of the project combined this approach with an appeal to public emotions by invoking patriotic sentiments in support of the dam's construction [113,114]. Aitken [115] shows how, in a public inquiry[5] on a Scottish windfarm, expertise of witnesses was highlighted by the developers' lawyer, "typically [focusing] on discrediting those of the opposite side. Cross-examination was used to demonstrate that witnesses were less qualified or appropriately experienced than their counterparts who presented evidence on the same topic" [p. 258].

3.2.7. Using regulatory and power structures to restrict public opposition

This category concerns publications which discuss responses of RET actors geared toward restricting of public opposition involving processes and actors from the institutional domain. Some developers and governments used legislation to bypass opponents and restrict their opportunities for opposition through formal decision-making [55,56,75,114–117], for example excluding opponents via a formal Environmental Impact Assessment procedure [118]. In a less explicit way, Hindmarsh [37] found that governmental policy responses to public opposition toward wind farms "appear more about empowering wind farm development than also effectively addressing the important issues raised by those directly vulnerable to the adverse impacts of wind farms" [p. 560].

Dunlap [119–121] describes practices of developers, politicians and what he calls 'elites' in the context of wind energy development in Mexico to include several forms of deception, coercion, intimidation and unequal benefit-sharing. This also involved long term avoidance of large-scale public consultation by regional politicians and elites [119]. In addition, "Land control and wind energy development (…) was largely executed through force by various state and extra-judicial forces, employing diplomatic and counterinsurgency techniques" [[119], p. 11]. Huber and Joshi [122] show a similar case in which national governmental actors and private developers attempt to restrict opposition toward hydropower projects in the Eastern Himalayas with authoritarian and coercive practices. Opponents were reportedly "victimized or threatened to be victimized, both discursively through a distinct anti-protest narrative, and materially by withholding patronage" [p. 19]. In both this case and the cases discussed by Siamanta and Dunlap [112], national governments are seen to restrict basic democratic rights to counter opposition. This is characteristic for a practice that Siamanta and Dunlap [112] label as 'green grabbing': "land and resources grabbing under an environmental ethic and rational" in Crete, Greece and Oaxaca, Mexico [p. 926].

3.2.8. Accommodating public concerns to reduce opposition

Several articles address ways in which RET actors try to address and accommodate concerns of residents or opponents by making changes to renewable energy projects, for example in design or location [47,79,123]. Jolivet and Heiskanen show French policymakers and project managers attempt to accommodate communities opposing wind

[5] The project developers appealed the refusal of the planning application by the local authorities. This appeal brought about a public inquiry.

energy by involving outside experts without ties to the developer and, eventually, making changes to the proposed projects to solve the controversies around size and visual impact [47]. Most articles, however, focus specifically on governmental actors and their attempts to address public concerns in the energy domain [10,124]. Other authors report on governments trying to accommodate concerns, specifically via changes in policies [125]. Lintz and Leibenath [126] also found evidence of a German state changing its policy to accommodate public concerns: "the protests (along with other factors) significantly contributed to the readjustment by the Saxon coalition government of its energy and climate policy and thus to a slowdown of energy-related landscape change" [p. 1]. And in some cases, RET actors postpone or cancel projects in the face of, or in response to, opposition [58,127–130], or withdraw financial support from controversial projects [131].

4. Discussion

Section 3 provides an overview of existing empirical social science research on RET actors in controversies surrounding renewable energy technologies and infrastructure. To answer our research question, we now know that this body of literature mostly focusses on two main categories. The first category (Section 3.1) concerns research on RET actors' perceptions of public opposition. These perceptions relate to the nature, impact and causes of public opposition, including specific groups of people and processes of engagement and decision-making. The second category (Section 3.2) describes research on different responses and strategies of RET actors to public opposition, which serve different purposes, including preventing, reducing and delegitimizing public opposition. On their own, these categories are helpful as they provide an overview of the type of research that has been conducted. Reviewed together, there are several observations to be made about this body of literature. Below, we discuss three characteristics regarding the diversity of the investigated actors, technologies and geography, the general descriptive nature of RET actor research and the type of practices that is mostly reported on. We end this section by discussing the implications of the reviewed literature for the broader context of research on energy controversies.

4.1. Little diversity in social science research on RET actors

Our methodological choice to only include English publications can be of influence on the diversity of the discussed publications: for example, had we included publications in other languages, we might have seen differences in the investigated countries, technologies, RET actors and journals. That being said, the included English publications do not reflect the diverse character of the renewable energy domain in terms of investigated actors, renewable energy technologies and infrastructure, and geography. The definition of RET actors that is used in this article describes a diverse group of actors in diverse organizations working on the implementation of renewable energy. However, in the discussed publications RET actors are often presented as a homogeneous and uniform group with similar interests, goals and discourses. Aside from the terms "developer" or "government", relatively little distinction is made within the RET actor category, the roles these actors take on or how they act in the context of public opposition (see also Fig. 1). There is little distinction between actors representing different departments (technical, legal, stakeholder engagement) of organizations developing projects, different types of organizations (public, private, etc.) or consortia consisting of different organizations.

The discussed publications also show little diversity in terms of the investigated renewable energy technologies and infrastructure: up till now the dominant focus has been wind energy, with little attention for other renewable energy technologies or infrastructure (Fig. 2). The striking difference between the number of publications on wind and other technologies, suggests that wind is more contentious. However, recent years have seen growing local opposition toward (plans for) solar projects. As such, this difference could simply be explained by the sheer amount of global installed wind capacity (564GW in 2018 [132]), as well as the relatively large number of (controversial) wind farms available for investigation. The geographical focus of the publications is also characterized by little diversity: five individual countries are investigated more often than five times (Fig. 3). Well-established research groups either located in or with a geographical preference for these countries could explain these higher numbers. Or perhaps it is related to what Dai [93] points out: "issues around public participation in energy transitions in non-Western democratic contexts have been an under-researched area" [p. 53].

Most articles are found in journals related to policy and planning (Fig. 4), which in itself is not surprising. What is notable is the limited presence of business, development or management-oriented journals. For example, we expected to find more research revolving around concepts such as social licence to operate (SLO) and corporate social responsibility (CSR). These concepts are widely researched in the non-renewable energy industry and the extractive industry (for example [133–135]). SLO and CSR practices in the renewable energy industry are investigated to a much lesser degree [135].

Future research needs to diversify in terms of renewable energy technologies and geographical contexts. This also opens up opportunities for comparative research: the publications included in this review offered a sample that was too small to make a meaningful comparison for example of the differences in technologies in relation to public opposition and the responses of RET actors. Future research could investigate questions such as: do RET actors' practices differ between different types of organizations (multinationals with no local connections, local energy cooperatives, etc.), energy technologies or geographical contexts? Comparison of RET actors' practices in projects with and without public opposition as well as comparison of RET actors with non-renewable energy actors are also interesting research avenues. Little is also known about the development of practices and assumptions over time: to what extent do RET actors (try to) learn from public opposition? Are there changes in their practices regarding (the design of) public participation or perspectives on the public and public opposition? Such research requires methods for comparative and longitudinal case studies.

In order to unpack the diversity of RET actors and their practices in the context of controversies, we need to gain a better understanding of who these RET actors and the organizations they represent actually are. Projects are often developed by different types of consortia (for example, public-private) where different interests and values meet. Which departments are involved and what kind of decision-making power do they have regarding (interaction with) public opposition? What are RET actors' discourses, assumptions, expectations and practices in the context of energy controversies and public opposition? What are their approaches to the planning and development of renewable energy? Such questions help build a more fine-grained and rich understanding of RET actors that goes beyond current generic descriptions and classifications such as "project developer" or "government".

4.2. Descriptive nature of RET actor literature

What is also notable is that the majority of publications has a descriptive nature: actions, practices or discourses of RET actors in response to, or in anticipation of, public opposition are described, but the majority of results do not provide an explanation. For example, Landeta-Manzano et al. [83] describe industry practices that are designed to achieve community acceptance of wind farms and simultaneously avoid conflict. However, in this publication, motivations or rationales underlying these practices, or explanations of why these practices are successful (or not), are not investigated. A reason for publications' descriptive nature might be that the majority of them concern case studies in which controversies are investigated from the perspective of multiple actors involved, including RET actors (see

Fig. 1). These multi-actor analyses often also have a strong focus on local communities or opponents and their experience of the controversy, while briefly reporting on RET actors. This means there is little room for in-depth investigation of RET actors and rationales underlying their responses to public opposition.

As such, there is a need for more explanatory research on RET actors and their practices. RET actors make assumptions and have expectations about the public, opponents, and their responses to renewable energy technologies and the planning process, which also shape RET actors' practices in terms of: "how they ought to operate, how they do operate, and how they can and should engage with the public" [[17], p.7]. Explanatory research can provide insights into how these actions and interactions of RET actors in energy controversies come about. What rationales do different types of organizations (public, private) have for their approach to public opposition, and how do these shape public engagement practices [17,136,137]? Research that explores relationships between different categories of RET actors and develops an understanding of networks of RET actors and interactions between them is also lacking.

Disciplines such as organizational studies, organizational anthropology and business administration and methods such as ethnography [138–140] can facilitate an explanatory understanding at inter- and intra-organizational levels. Questions related to organizational culture, identity and power dynamics [141,142] can help gain insights into intra-organizational dynamics and subsequently into explanations of (engagement) practices and interactions in the context of public opposition. For example, what kind of intra-organizational constraints and enablers do they experience when public opposition is anticipated or encountered [44]? How do organizational disagreements about (the level of) accommodation of public interests and concerns affect decisions on engagement with opposition?

4.3. Little research on RET actors accommodating public concerns

Most of the identified responses show RET actors trying to avoid or reduce public opposition. Or in a more direct way, they attempt to oppress or restrict it. Section 3.2.8 provides some examples of RET actors adapting projects or processes to accommodate public concerns. Nevertheless, this review has unearthed little research that investigates this type of responsiveness of RET actors toward public opposition in controversies. This is interesting, as there are more and more pleas for learning from public opposition and approaching it as a source of information about the diversity of normative appraisals in renewable energy planning [18,30,143]. These pleas are coming from academia, as well as practitioners. One example of RET actors attempting to engage with public opposition in a responsive way, that we know of from the Dutch context, is the Energy and Environment Learning Platform.[6] This community of practice, initiated by the energy industry and the Ministry of Economic Affairs and Climate, is geared toward knowledge exchange on community engagement in the energy transition. Other examples are voluntary codes of conduct on stakeholder engagement in renewable energy development, like the one created by the Dutch Association of Geothermal Operators [144]. However, such initiatives and practices of RET actors are not reported in this literature review. Perhaps the relatively slow speed of scientific publications is still catching up with such evolving practices in the world of energy transitions.

4.4. Implications for the broader context of research on energy controversies

Following the results discussed above, we share two observations that shed more light on the investigation of RET actors in the broader field of renewable energy controversies and its conflict dynamics.

The first observation we made while conducting this literature review is that some researchers explicitly highlight a bias in their research on RET actors [43,80]. For example, Hall et al. [80] use documents from wind energy developers to:

> understand the wind industry's current thinking, practices and trends in community engagement and benefit sharing. While we recognise a bias in this perspective, and note that research has recommended the benefits of holistic insights from multiple stakeholder perspectives and experiences [43], these strategic documents are rarely shared externally and provide a unique developer-centric perspective that is rarely accessible [p. 125].

Although Hall et al. do explicitly state the value of this developer-centric approach, it is preceded by a disclaimer. One could ask why developer-centric research is explicitly labeled as biased, while most research on opponents, communities and residents engaged in energy controversies does not explicitly receive such a label. This creates the impression that investigation of certain actors is somehow equal to being their ally or championing their interests. We should follow the lead of Burningham et al. [19], Cass and Walker [59], and Hall [74], and be cautious "not to vilify developers or to romanticize project opponents (Lake 1993, Wexler 1996)" [[19], p. 258].

The second observation was made while we determined the eligibility of publications for this literature review. During this process, we noticed quite a number of articles and chapters that investigated practices of RET actors through the eyes of other stakeholders. Often these publications concerned stakeholders who were affected by RET actors' practices, such as residents and opponents [48,55,56,114,116–118,120,121,130,131,145]. Researchers described and discussed these practices in more depth through accounts of other actors and their experiences with these practices. Often, these analyses "through the eyes of others" discuss practices that are characterized as having harmful consequences for local communities and opponents. Though several of these publications were out of scope for this literature review, we do feel it is important to share this observation, as it illustrates the aforementioned "obsession" of social science researchers with the public [28] and perhaps also hints at the tendencies to vilify RET actors and romanticize opponents [19].

The above observations point to two 'tendencies' of researchers investigating RET actors: presenting RET actor research as biased and investigating RET actors through the perspective of other stakeholders. These tendencies could be born out of necessity and pragmatism: several researchers [43,56,130] report that developers were not willing to participate in research projects due to existing tensions in controversies, or were only willing to respond via a written statement. Getting access to RET actors is thus an important challenge for social science researchers.

The two tendencies could also indicate potential implicit assumptions among researchers regarding which perspectives, and subsequently which information, is considered relevant and legitimate when investigating energy controversies. This is potentially problematic as such research often serves as a starting point for recommendations to RET actors: what they should and should not do from the perspective of the affected or opposing stakeholders, or according to the researchers (for example [48,63,97,145]). As a research field that prides itself on a critical and reflective attitude, it might be insightful to apply this attitude to our own assumptions regarding RET actors.

One could argue that RET actors are generally in a more powerful position compared to communities and those opposing projects, and therefore the latter are often provided with a voice by researchers as a type of counterhegemonic action. Such an approach is justifiable as a way of addressing power asymmetries. At the same time, the importance of investigating powerful actors and elites has long been pointed out [146,147]. By "obsessing" over publics [28], we run the risk of missing or misinterpreting potential valuable insights when investigating renewable energy controversies. We may very well come to different conclusions or recommendations if we opened up our research attention

[6] LEO, in Dutch *Lerend Platform Energie en Omgeving*, www.platformleo.nl.

to include RET actors – especially given our finding that RET actors are treated with low resolution, i.e., as a rather homogeneous group, without much attention to diversity among RET actors. As a field, we need to have discussions on how to investigate multiple actors and perspectives, without compromising our ability to address apparent power asymmetries.

5. Conclusion

With this literature review we set out to map the current state of knowledge on RET actors in controversies surrounding renewable energy technologies and infrastructure. Reviewing the existing empirical social science research on RET actors reveals two main categories in this body of literature: RET actors' perceptions of public opposition (Section 3.1) and their responses to public opposition (Section 3.2). Reflecting on the literature, we made three observations (Section 4): (1) The diversity of RET actors, types of energy technologies or geographical areas is not reflected in the body of literature; (2) Publications are mainly of a descriptive character; (3) there is little research on how RET actors (attempt to) accommodate public concerns. Reflecting on the reviewed literature in the context of the broader field reveals that research that does focus on RET actors is on multiple occasions presented as biased. In addition, it often investigates RET actors through the eyes of others, emphasizing a counterhegemonic perspective on renewable energy controversies.

With this review, we add to the current literature by providing a substantiation for existing calls [19,42,44,45] to broaden the scope of social science research. In the discussion (Section 4), we have provided a number of suggestions for future research. These suggestions translate into an agenda for future social science research into RET actors involved in energy controversies:

1. The reviewed literature (with a few exceptions) currently shows a one-dimensional, generic picture of RET actors, their perspectives and practices. As such, this diversity of RET actors and their practices in the context of controversies needs to be unpacked. Future research also needs to diversify in terms of renewable energy technologies and geographical contexts. This also opens up opportunities for comparative research.
2. Due to the mainly descriptive character research into RET actors, future research needs to be aimed at providing explanations for RET actors' perspectives and practices. Such explanatory research helps build a more fine-grained and rich understanding of RET actors that goes beyond current generic descriptions.

Besides these two general directions, we want to make two additional points resulting from our research design. First, as mentioned earlier, our methodological choice to only include publications in English means we are potentially missing valuable insights into RET actors in the context of controversies published in other languages. As such, future research should include publications in other languages. Second, as we reviewed RET actors in the context of societal conflict means that related studies on this group which do not explicitly focus on societal conflict were not included. Such publications could provide valuable insights in terms of practices regarding engagement with (local) publics and communities (for example [134,148]). Reviewing RET actors, their perspectives and practices without societal conflict as a demarcation would provide a more comprehensive picture.

There is a clear need to further unpack the metaphorical black box of RET actors. Such research can contribute to a more fine-grained understanding of the (development of) conflict dynamics between the actors engaged in controversies and action perspectives for the involved actors. These insights would allow researchers to identify challenges and opportunities in the governance of renewable energy controversies, and provide recommendations for just and democratic energy transitions that are more attuned to RET actors. This would be both timely and necessary considering the continuation and upscaling of energy transitions worldwide.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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Data availability

Data will be made available on request.

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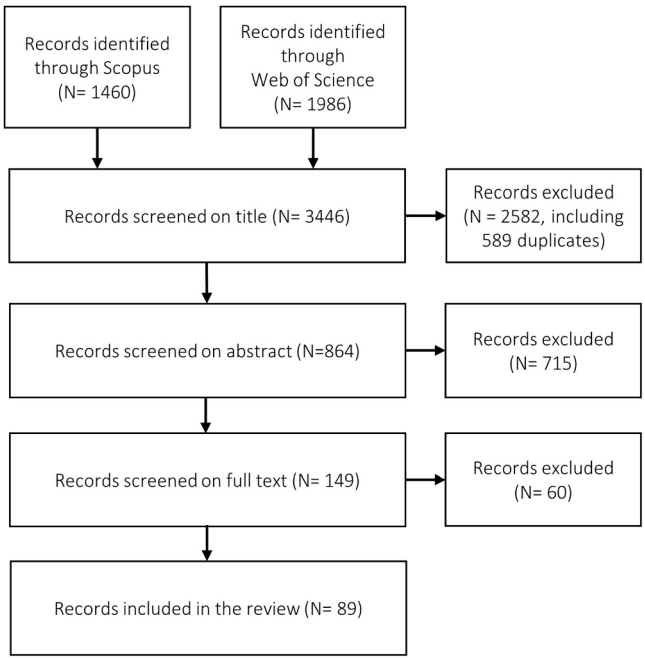
Appendix A. Flow chart of selection publications for systematic literature review



Appendix B. Overview publications included in literature review

Publications (in alphabetical order)	RET actors studied	Renewable energy technology & infrastructure	Geographical focus	Theme
M. Aguilar-Støen, C. Hirsch, Bottom-up responses to environmental and social impact assessments: A case study from Guatemala, Environ. Impact Assess. Rev. 62 (2017)	Multiple actors, including RET actors	Hydropower	Guatemala	3.2.1
M. Aguilar-Støen, C. Hirsch, Environmental Impact Assessments, local power and self-determination: The case of mining and hydropower development in Guatemala, Extr. Ind. Soc. 2 (2015) 472–479.	Multiple actors, including RET actors	Hydropower	Guatemala	3.2.7
M. Aitken, A three-dimensional view of public participation in Scottish land-use planning: Empowerment or social control?, Plan. Theory. 9 (2010) 248–264	Multiple actors, including RET actors	Onshore wind	Scotland	3.2.6; 3.2.7
M. Aitken, Wind power and community benefits: Challenges and opportunities, Energy Policy. 38 (2010) 6066–6075.	Multiple actors, including RET actors	Onshore wind	United Kingdom	3.2.5
C. Anderson, The networked minority: How a small group prevailed in a local windfarm conflict, Energy Policy. 58 (2013) 97–108.	Multiple actors, including RET actors	Onshore wind	Australia	3.2.1
E. Atkins, Dammed and diversionary: The multi-dimensional framing of Brazil's Belo Monte dam, Singap. J. Trop. Geogr. 38 (2017) 276–292.	Government - national	Hydropower	Brazil	3.2.6
J. Barry, G. Ellis, C. Robinson, Cool rationalities and hot air: A rhetorical approach to understanding debates on renewable energy, Glob. Environ. Polit. 8 (2008) 67–98.	RET actors and opponents	Onshore wind	Northern Ireland	3.1.3; 3.2.4; 3.2.6
P. Bosley, K. Bosley, Public acceptability of California's wind energy developments: Three studies, Wind Eng. 12 (1988) 311–318.	Multiple actors, including RET actors	Onshore wind	United States	3.1.2
P. Boucher, Biofuel development in the UK: Regulatory and engineering visions beyond a changing controversy, in: T. Roberts, P. Upham, C. Mclachlan, S. Mander, C. Gough, P. Boucher, D. Abi Ghanem (Eds.), Low-Carbon Energy Controv., 1st ed., Routledge, London, 2013: pp. 177–196.	Multiple actors, including RET actors	Bioenergy	United Kingdom	3.2.8
S. Bourdin, M. Colas, F. Raulin, Understanding the problems of biogas production deployment in different regions: territorial governance matters too, J. Environ. Plan. Manag. 63 (2020) 1655–1673.	Multiple actors, including RET actors	Bioenergy	France	3.2.1
S. Breukers, M. Wolsink, Wind power implementation in changing institutional landscapes: An international comparison, Energy Policy. 35 (2007) 2737–2750	Multiple actors, including RET actors	Onshore wind	Multiple: England, Germany, The Netherlands	3.1.3
A. Bues, Planning, Protest, and Contentious Politics: The Governance of Wind Energy in Brandenburg and Ontario, Disp - Plan. Rev. 54 (2018) 34–45.	RET actors and opponents	Wind	Multiple: Canada, Germany	3.2.3
K. Burningham, J. Barnett, G. Walker, An Array of Deficits: Unpacking NIMBY Discourses in Wind Energy Developers' Conceptualizations of Their Local Opponents, Soc. Nat. Resour. 28 (2015) 246–260.	Multiple - industry	Multiple RET	United Kingdom	3.1.3; 3.2.4; 3.2.6
N. Cass, G. Walker, Emotion and rationality: The characterisation and evaluation of opposition to renewable energy projects, Emot. Sp. Soc. 2 (2009) 62–69.	Multiple actors, including RET actors	Wind	United Kingdom	3.1.1; 3.1.2; 3.2.3; 3.2.4

(*continued on next page*)

(*continued*)

Publications (in alphabetical order)	RET actors studied	Renewable energy technology & infrastructure	Geographical focus	Theme
T. Christidis, G. Lewis, P. Bigelow, Understanding support and opposition to wind turbine development in Ontario, Canada and assessing possible steps for future development, Renew. Energy. 112 (2017) 93–103.	Multiple actors, including RET actors	Wind	Canada	3.1.2
M. Cotton, P. Devine-Wright, Discourses of energy infrastructure development: A Q-method study of electricity transmission line siting in the UK, Environ. Plan. A. 43 (2011) 942–960.	Multiple actors, including RET actors	Electricity transmission lines	United Kingdom	3.1.2
M. Cotton, P. Devine-Wright, NIMBYism and community consultation in electricity transmission network planning, in: P. Devine-Wright (Ed.), Renew. Energy Public From NIMBY to Particip., Earthscan, Routledge, London, 2011: pp. 115–128.	Electricity network operators	Electricity transmission lines	United Kingdom	3.1.2; 3.2.1
R. Cowell, Wind power and "the planning problem": The experience of Wales, Eur. Environ. 17 (2007) 291–306.	Government - national	Onshore wind	Wales	3.2.8
R. Cowell, G. Bristow, M. Munday, Acceptance, acceptability and environmental justice: The role of community benefits in wind energy development, 0568 (2011).	Multiple actors, including RET actors	Onshore wind	Wales	3.2.5
Y. Dai, Policy instrument designed to gain transition legitimacy: A case of Chinese nuclear development, Environ. Innov. Soc. Transitions. 30 (2019) 43–58.	Multiple RET actors	Nuclear energy	China	3.2.2 3.2.4
J. Dennison, Entangled sovereignties: The Osage Nation's interconnections with governmental and corporate authorities, Am. Ethnol. 44 (2017) 684–696.	Multiple actors, including RET actors	Onshore wind	United States	3.2.7
P. Díaz, C. Adler, A. Patt, Do stakeholders' perspectives on renewable energy infrastructure pose a risk to energy policy implementation? A case of a hydropower plant in Switzerland, Energy Policy. 108 (2017) 21–28.	RET actors and opponents	Hydropower	Switzerland	3.1.2
A. Dunlap, "A Bureaucratic Trap:" Free, Prior and Informed Consent (FPIC) and Wind Energy Development in Juchitán, Mexico, Capital. Nature, Social. 29 (2018) 88–108.	Multiple actors, including RET actors	Onshore wind	Mexico	3.2.7
A. Dunlap, Counterinsurgency for wind energy: the Bíi Hioxo wind park in Juchitán, Mexico, J. Peasant Stud. 45 (2018) 630–652.	Multiple actors, including RET actors	Onshore wind	Mexico	3.2.7
A. Dunlap, Insurrection for land, sea and dignity: Resistance and autonomy against wind energy in Álvaro Obregón, Mexico, J. Polit. Ecol. 25 (2018) 120–143.	Multiple actors, including RET actors	Onshore wind	Mexico	3.2.7
A. Dunlap, The Politics of Ecocide, Genocide and Megaprojects: Interrogating Natural Resource Extraction, Identity and the Normalization of Erasure, J. Genocide Res. 0 (2020) 1–24.	Multiple actors, including RET actors	Onshore wind	Mexico	3.2.7
S. Fast, Qualified, absolute, idealistic, impatient: dimensions of host community responses to wind energy projects, Environ. Plan. A. 47 (2015) 1540–1557.	Multiple actors, including RET actors	Onshore wind	Canada	3.1.3
C.A. Frate, C. Brannstrom, Stakeholder subjectivities regarding barriers and drivers to the introduction of utility-scale solar photovoltaic power in Brazil, Energy Policy. 111 (2017) 346–352.	Multiple - industry	Solar	Brazil	3.1.2
T. Gray, C. Haggett, D. Bell, Offshore wind farms and commercial fisheries in the UK: A study in stakeholder consultation, Ethics, Place Environ. 8 (2005) 127–140.	Multiple actors, including RET actors	Offshore wind	United Kingdom	3.2.5
N. Hall, P. Ashworth, P. Devine-Wright, Societal acceptance of wind farms: Analysis of four common themes across Australian case studies, Energy Policy. 58 (2013) 200–208.	Multiple actors, including RET actors	Onshore wind	Australia	3.1.3
N.L. Hall, J. Hicks, T. Lane, E. Wood, Planning to engage the community on renewables: insights from community engagement plans of the Australian wind industry, Australas. J. Environ. Manag. 27 (2020) 123–136.	Developers	Onshore wind	Australia	3.2.1
N. L. Hall, The discourse of "social licence to operate": case study of the Australian wind industry, AIMS Energy. 2 (2014) 443–460.	Stakeholder engagement professionals	Onshore wind	Australia	3.1.3; 3.2.1; 3.2.5
R. Hindmarsh, Wind Farms and Community Engagement in Australia: A Critical Analysis for Policy Learning, East Asian Sci. Technol. Soc. 4 (2010) 541–563.	Government - sub-national	Onshore wind	Australia	3.1.3; 3.2.4; 3.2.6; 3.2.7
R. Hindmarsh, Hot air ablowin! "Media-speak", social conflict, and the Australian "decoupled" wind farm controversy, Soc. Stud. Sci. 44 (2014) 194–217.	Multiple actors, including RET actors	Onshore wind	Australia	3.2.6
C. Howe, Ecoautoridad Antropocenica: Los Vientos de Oaxaca, Anthropol. Q. 87 (2014) 381–404.	Multiple actors, including RET actors	Onshore wind	Mexico	3.2.6
A. Huber, D. Joshi, Hydropower, Anti-Politics, and the Opening of New Political Spaces in the Eastern Himalayas, World Dev. 76 (2015) 13–25.	Multiple actors, including RET actors	Hydropower	India	3.2.7
A. Israel, R.J. Herrera, The governance of Peruvian energy transitions: Path dependence, alternative ideas and change in national hydropower expansion, Energy Res. Soc. Sci. 69 (2020) 101608.	Multiple actors, including RET actors	Hydropower	Peru	3.2.1
A.A. Jami, P.R. Walsh, From consultation to collaboration: A participatory framework for positive community engagement with wind energy projects in Ontario, Canada, Energy Res. Soc. Sci. 27 (2017) 14–24.	Multiple actors, including RET actors	Onshore wind	Canada	3.1.1; 3.1.3
A.A. Jami, P.R. Walsh, Wind power deployment: The role of public participation in the decision-making process in Ontario, Canada, Sustain. 8 (2016).	Multiple actors, including RET actors	Onshore wind	Canada	3.2.2
T. Jenssen, The good, the bad, and the ugly: Acceptance and opposition as keys to bioenergy technologies, J. Urban Technol. 17 (2010) 99–115.	Multiple actors, including RET actors	Bioenergy	Germany	3.1.3

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Publications (in alphabetical order)	RET actors studied	Renewable energy technology & infrastructure	Geographical focus	Theme
D. Jijelava, F. Vanclay, How a large project was halted by the lack of a social Licence to operate: Testing the applicability of the Thomson and Boutilier model, Environ. Impact Assess. Rev. 73 (2018) 31–40.	Multiple actors, including RET actors	Hydropower	Georgia	3.2.5; 3.2.7
E. Jolivet, E. Heiskanen, Blowing against the wind-An exploratory application of actor network theory to the analysis of local controversies and participation processes in wind energy, Energy Policy. 38 (2010) 6746–6754.	Multiple actors, including RET actors	Onshore wind	France	3.2.8
N. Juerges, J. Leahy, J. Newig, A typology of actors and their strategies in multi-scale governance of wind turbine conflict within forests, Land Use Policy. 96 (2020) 104691.	Multiple actors, including RET actors	Onshore wind	Multiple: Germany, United States	3.2.5
N. Juerges, J. Newig, What role for frames in scalar conflicts?, Land Use Policy. 49 (2015) 426–434.	Multiple actors, including RET actors	Onshore wind	Germany	3.1.2; 3.1.3
S. Kerr, K. Johnson, S. Weir, Understanding community benefit payments from renewable energy development, Energy Policy. 105 (2017) 202–211.	Multiple actors, including RET actors	Multiple RET	United Kingdom	3.2.4
J. Kirchherr, K. J. Charles, M.J. Walton, The interplay of activists and dam developers: the case of Myanmar's mega-dams, Int. J. Water Resour. Dev. 33 (2017) 111–131.	RET actors and opponents	Hydropower	Myanmar	3.2.8
B. Landeta-Manzano, G. Arana-Landín, P.M. Calvo, I. Heras-Saizarbitoria, Wind energy and local communities: A manufacturer's efforts to gain acceptance, Energy Policy. 121 (2018) 314–324.	Developers	Onshore wind	Multiple	3.2.1; 3.2.5
R. Lawrence, Internal colonisation and Indigenous resource sovereignty: Wind power developments on traditional Saami lands, Environ. Plan. D Soc. Sp. 32 (2014) 1036–1053.	Multiple actors, including RET actors	Onshore wind	Sweden	3.2.7
M. Leibenath, A. Otto, Competing Wind Energy Discourses, Contested Landscapes, Landsc. Online. 38 (2014) 1–18.	RET actors and opponents	Onshore wind	Germany	3.2.6
G. Lintz, M. Leibenath, The politics of energy landscapes: The influence of local anti-wind initiatives on state policies in Saxony, Germany, Energy. Sustain. Soc. 10 (2020).	RET actors and opponents	Onshore wind	Germany	3.2.8
M.È. Maillé, J. Saint-Charles, Fuelling an environmental conflict through information diffusion strategies, Environ. Commun. 8 (2014) 305–325.	RET actors and opponents	Onshore wind	Canada	3.2.7
N. Martinez, N. Komendantova, The effectiveness of the social impact assessment (SIA) in energy transition management: Stakeholders' insights from renewable energy projects in Mexico, Energy Policy. 145 (2020) 111744.	Multiple actors, including RET actors	Multiple RET	Mexico	3.2.2; 3.2.3
E. Martínez-Mendoza, L.A. Rivas-Tovar, L.E. García-Santamaría, Wind energy in the Isthmus of Tehuantepec: conflicts and social implications, Environ. Dev. Sustain. (2021).	Multiple actors, including RET actors	Onshore wind	Mexico	3.1.2
A. McClanahan, The ethics of landscape: Discourses of cultural and environmental sustainability in the heart of Neolithic Orkney World Heritage Site, in: L. Bourdeau, M. Gravari-Barbas, M. Robinson (Eds.), World Heritage, Tour. Identity Inscr. Co-Production, Routledge, 2015: pp. 217–226.	Multiple actors, including RET actors	Onshore wind	Scotland	3.2.5
C. McLachlan, S. Mander, What have facts got to do with it anyway? Competing knowledge claims in low-carbon energy controversy, in: T. Roberts, P. Upham, C. Mclachlan, S. Mander, C. Gough, P. Boucher, D. Abi Ghanem (Eds.), Low-Carbon Energy Controv., 1st ed., Routledge, London, 2013: pp. 85–113.	Multiple actors, including RET actors	Multiple RET	United Kingdom	3.1.3; 3.2.8
S. Mjahed Hammami, S. Chtourou, H. Al Moosa, A holistic approach to understanding the acceptance of a community-based renewable energy project: A pathway to sustainability for Tunisia's rural region, Bus. Strateg. Environ. 27 (2018) 1535–1545.	Multiple actors, including RET actors	Onshore wind	Tunisia	3.2.5
K. Mogensen, From public relations to corporate public diplomacy, Public Relat. Rev. 43 (2017) 605–614.	Multiple actors, including RET actors	Hydropower	Myanmar	3.2.8
K.K. Mulvaney, P. Woodson, L.S. Prokopy, A tale of three counties: Understanding wind development in the rural Midwestern United States, Energy Policy. 56 (2013) 322–330.	RET actors and opponents	Onshore wind	United States	3.1.3
T. Niazi, Contesting Instrumental Knowledge With Communicative Action: Why Kalabagh Dam (Pakistan) Remains Unbuilt, Organ. Environ. 32 (2019) 441–465.	Multiple actors, including RET actors	Hydropower	Pakistan	3.2.6
M.J. Pasqualetti, C. Schwartz, Siting Solar Power in Arizona: A Public Value Failure?, in: P. Devine-Wright (Ed.), Renew. Energy Public From NIMBY to Particip., 1ste ed., Routledge, London, 2011: pp. 167–185.	Multiple actors, including RET actors	Onshore wind	United States	3.2.2
M.J. Pasqualetti, Wind energy landscapes: Society and technology in the California desert, Soc. Nat. Resour. 14 (2001) 689–699.	Multiple actors, including RET actors	Solar	United States	3.2.4
Y. Pepermans, I. Loots, Wind farm struggles in Flanders fields: A sociological perspective, Energy Policy. 59 (2013) 321–328.	Multiple actors, including RET actors	Onshore wind	Belgium	3.2.2; 3.2.3; 3.2.4
A. Proka, M. Hisschemöller, D. Loorbach, Transition without conflict? Renewable energy initiatives in the Dutch energy transition, Sustain. 10 (2018).	Multiple RET actors	Multiple RET	The Netherlands	3.1.3; 3.2.3
J. Ramirez, Contentious Dynamics Within the Social Turbulence of Environmental (In)justice Surrounding Wind Energy Farms in Oaxaca, Mexico, J. Bus. Ethics. 169 (2021) 387–404.	Multiple actors, including RET actors	Onshore wind	Mexico	3.2.8
J. Ramirez, Indigenous communities and mega-projects: Corporate social responsibility (CSR) and consultation-consent principles, in: D. Blowfield,	Multiple actors, including RET actors	Onshore wind	Mexico	3.2.8

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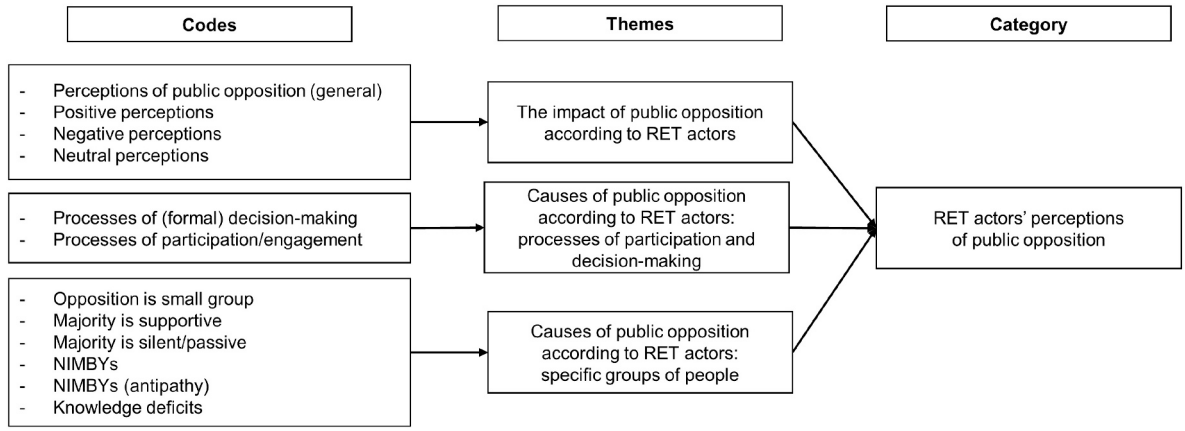
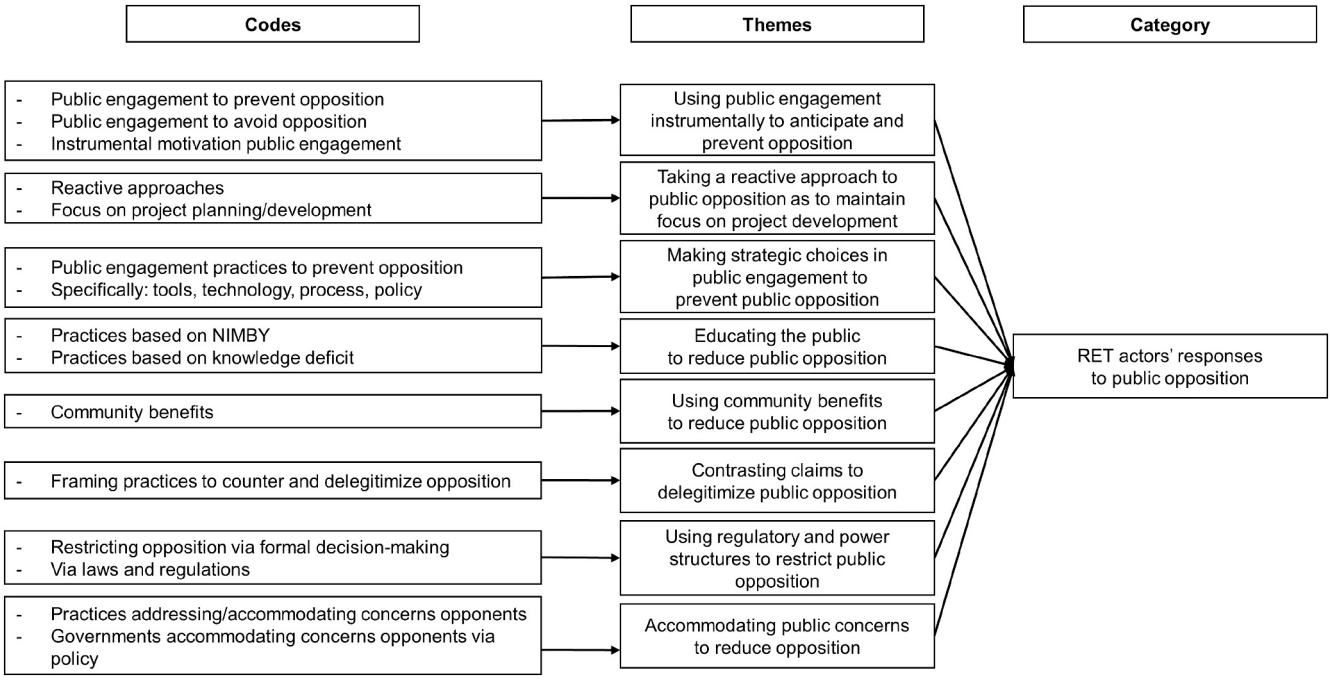
Publications (in alphabetical order)	RET actors studied	Renewable energy technology & infrastructure	Geographical focus	Theme
M., Karam, C., & Jamali (Ed.), Dev. Corp. Soc. Responsib. Vol. 1 Multinatl. Corp. Glob. Context, 1st editio, Routledge, 2017.				
K.E. Rignall, Solar power, state power, and the politics of energy transition in pre-Saharan Morocco, Environ. Plan. A. 48 (2016) 540–557.	Government - national	Solar	Morocco	3.2.6
D. Rudolph, J.K. Kirkegaard, Making Space for Wind Farms: Practices of Territorial Stigmatisation in Rural Denmark, Antipode. 51 (2019) 642–663.	Developers	Onshore wind	Denmark	3.2.1
I.L. Saglie, T.H. Inderberg, H. Rognstad, What shapes municipalities' perceptions of fairness in windpower developments?, Local Environ. 25 (2020) 147–161.	Government - local	Onshore wind	Norway	3.2.8
P. Scherhaufer, S. Höltinger, B. Salak, T. Schauppenlehner, J. Schmidt, Patterns of acceptance and non-acceptance within energy landscapes: A case study on wind energy expansion in Austria, Energy Policy. 109 (2017) 863–870.	Multiple actors, including RET actors	Onshore wind	Austria	3.1.3; 3.2.3; 3.2.6
Z.C. Siamanta, A. Dunlap, "Accumulation by wind energy": Wind energy development as a capitalist Trojan horse in Crete, Greece and Oaxaca, Mexico, Acme. 18 (2019) 925–955.	Multiple actors, including RET actors	Onshore wind	Multiple: Greece, Mexico	3.2.6; 3.2.7
R.G. Sierra, A.Z. Sarmiento, Hydropower megaprojects in Colombia and the influence of local communities: A view from prospect theory to decision making process based on expert judgment used in large organizations, Int. J. Energy Econ. Policy. 6 (2016) 408–420.	Multiple actors, including RET actors	Hydropower	Colombia	3.2.1
N. Simcock, Procedural justice and the implementation of community wind energy projects: A case study from South Yorkshire, UK, Land Use Policy. 59 (2016) 467–477.	RET actors and opponents	Onshore wind	United Kingdom	3.1.2; 3.1.3; 3.2.1; 3.2.5
E. Songsore, M. Buzzelli, J. Baxter, Understanding developer perspectives and experiences of wind energy development in Ontario, Environ. Plan. C Polit. Sp. 36 (2018) 649–668.	Developers	Wind	Canada	3.1.1; 3.1.3; 3.2.5
T. Spiess, C. De Sousa, Barriers to Renewable Energy Development on Brownfields, J. Environ. Policy Plan. 18 (2016) 507–534.	Multiple - industry	Multiple RET	Multiple: majority participants United States	3.1.1; 3.2.4
E.R. Stafford, C.L. Hartman, Resolving community concerns over local wind power development in Utah, Sustainability. 5 (2012) 38–43.	Multiple actors, including RET actors	Onshore wind	United States	3.2.1
M. Suškevičs, S. Eiter, S. Martinat, D. Stober, E. Vollmer, C.L. de Boer, M. Buchecker, Regional variation in public acceptance of wind energy development in Europe: What are the roles of planning procedures and participation?, Land Use Policy. 81 (2019) 311–323.	Experts	Wind	Multiple: Europe	3.1.2
R. Tafon, D. Howarth, S. Griggs, The politics of Estonia's offshore wind energy programme: Discourse, power and marine spatial planning, Environ. Plan. C Polit. Sp. 37 (2019) 157–176.	Multiple actors, including RET actors	Offshore wind	Estonia	3.2.6
B.R. Upreti, Conflict over biomass energy development in the United Kingdom: Some observations and lessons from England and Wales, Energy Policy. 32 (2004) 785–800.	Multiple actors, including RET actors	Bioenergy	United Kingdom	3.2.4
E. van de Grift, E. Cuppen, S. Spruit, Co-creation, control or compliance? How Dutch community engagement professionals view their work, Energy Res. Soc. Sci. 60 (2020).	Stakeholder engagement professionals	Multiple RET	The Netherlands	3.2.1; 3.2.2
E.C. van der Waal, H.J. van der Windt, R. Botma, E.C.J. van Oost, Being a better neighbor: A value-based perspective on negotiating acceptability of locally-owned wind projects, Sustain. 12 (2020) 1–18.	Multiple actors, including RET actors	Onshore wind	The Netherlands	3.2.1
I. Verhoeven, Contentious governance around climate change measures in the Netherlands, Env. Polit. 00 (2020) 1–23.	RET actors and opponents	Nearshore wind	The Netherlands	3.2.6
Å. Waldo, Offshore wind power in Sweden-A qualitative analysis of attitudes with particular focus on opponents, Energy Policy. 41 (2012) 692–702.	Multiple actors, including RET actors	Offshore wind	Sweden	3.1.3; 3.2.4
C. Walker, J. Baxter, Procedural justice in Canadian wind energy development: A comparison of community-based and technocratic siting processes, Energy Res. Soc. Sci. 29 (2017) 160–169.	Multiple actors, including RET actors	Wind	Canada	3.2.1
C. Walker, J. Baxter, "It's easy to throw rocks at a corporation": wind energy development and distributive justice in Canada, J. Environ. Policy Plan. 19 (2017) 754–768.	Multiple actors, including RET actors	Onshore wind	Canada	3.2.5
G. Walker, N. Cass, K. Burningham, J. Barnett, Renewable energy and sociotechnical change: Imagined subjectivities of "the public" and their implications, Environ. Plan. A. 42 (2010) 931–947.	Multiple actors, including RET actors	Multiple RET	United Kingdom	3.1.1; 3.1.3; 3.2.1; 3.2.5; 3.2.8
S. Weir, S. Kerr, Property, power and planning: Attitudes to spatial enclosure in Scottish seas, Mar. Policy. 108 (2019) 103633.	Multiple actors, including RET actors	Multiple RET	Scotland	3.2.8
M. Wolsink, Discourses on the Implementation of Wind Power: Stakeholder Views on Public Engagement, in: P. Devine-Wright (Ed.), Renew. Energy Public From NIMBY to Particip., 1st ed., Routledge, London, 2014: pp. 75–87.	Multiple actors, including RET actors	Onshore wind	Multiple: England, Germany, The Netherlands	3.2.1
M. Wolsink, Near-shore wind power-Protected seascapes, environmentalists' attitudes, and the technocratic planning perspective, Land Use Policy. 27 (2010) 195–203.	Multiple actors, including RET actors	Nearshore wind	The Netherlands	3.1.3; 3.2.7
E. Zárate-Toledo, R. Patiño, J. Fraga, Justice, social exclusion and indigenous opposition: A case study of wind energy development on the Isthmus of Tehuantepec, Mexico, Energy Res. Soc. Sci. 54 (2019) 1–11.	Multiple actors, including RET actors	Onshore wind	Mexico	3.2.8

Appendix C. Overview of main codes per theme and category





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